

Table 1. RealToxicityPrompts results with evaluated-model KV cache disabled. Values are mean $\pm$ standard error across five repetitions; TBD marks unrun model-method pairs.

	Method	Toxic continuations (%) $\downarrow$	Distinct-2 $\uparrow$	Perplexity $\downarrow$
Gemma-2-2B	Original	3.96 ± 0.31	0.654 ± 0.002	10.98 ± 0.07
	ITI	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD
	S-PID	0.16 ± 0.06	0.682 ± 0.001	11.18 ± 0.04
	A-LQR	0.00 ± 0.00	0.630 ± 0.001	24.62 ± 0.17
	H_∞ (ours)	0.12 ± 0.02	0.683 ± 0.001	14.00 ± 0.08
Llama-3-8B	Original	TBD	TBD	TBD
	ITI	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD
Qwen-2.5-14B	Original	TBD	TBD	TBD
	ITI	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD